

TIANYU WANG

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ACADEMIC APPOINTMENTS

The Chinese University of Hong Kong

Assistant Professor, Systems Engineering and Engineering Management

Hong Kong SAR
2027 -

Harvard Business School

Postdoctoral Fellow, AI Institute

Boston, MA
2026 - 2027

EDUCATION

Columbia University

M.S. & Ph.D. in Operations Research

Advisors: Garud Iyengar, Henry Lam

New York, NY
2021 - 2026

Tsinghua University

B.E. in Information Systems

B.S. in Pure and Applied Mathematics

Beijing, China
2017 - 2021
2018 - 2021

National University of Singapore

Exchange Student

Singapore
2019

PUBLICATIONS AND WORKING PAPERS

Note. *: Authors are listed in alphabetical order. ⁺: Authors are equally contributed.

JOURNAL ARTICLES (INCLUDING MINOR REVISIONS)

1. Garud Iyengar*, Henry Lam*, Tianyu Wang*. Optimizer's Information Criterion: Dissecting and Correcting Bias in Data-Driven Optimization. *Minor revision at Management Science*. [\[Link\]](#)
 - **Honorable Mention of the Triennial Dupačová-Prékopa Best Student Paper Prize in Stochastic Programming 2025.**
2. Tianyu Wang⁺, Jiashuo Liu⁺, Peng Cui, Hongseok Namkoong. Rethinking Distribution Shifts: Empirical Analysis and Modeling for Tabular Data. *Management Science*, *forthcoming*. Preliminary version appeared in NeurIPS 2023. [\[Link\]](#) [\[Python Package\]](#)
3. Chi Seng Pun*, Tianyu Wang*, Zhenzhen Yan*. Data-Driven Distributionally Robust CVaR Portfolio Optimization Under A Regime-Switching Ambiguity Set. *Manufacturing & Service Operations Management*, 25(5):1779-1795, 2023. [\[Link\]](#)

REFEREED CONFERENCE PUBLICATIONS

1. Derek Long, Tianyu Wang, Henry Lam, Jay Sethuraman, Matthew Adams, Monish Dadlani, Kathleen Thomson, Ayten Turkcan et al. A Hybrid Simulation-Machine Learning Approach to Improve EMS Responses in New York City. *Winter Simulation Conference (WSC) 2026*.
2. Tianyu Wang, Luhao Zhang, Rachel Cummings. Integrating Feature Correlation in Differential Privacy with Applications in DP-ERM. *Artificial Intelligence and Statistics (AISTATS) 2026*.
3. Garud Iyengar*, Henry Lam*, Tianyu Wang*. Is Cross-Validation the Gold Standard to Estimate Out-of-sample Model Performance? *Neural Information Processing Systems (NeurIPS) 2024*. [\[Link\]](#)

4. Jiashuo Liu, Jiayun Wu, Tianyu Wang, Hao Zou, Peng Cui. Geometry-Calibrated DRO: Combating Over-Pessimism with Free Energy Implications. *International Conference on Machine Learning (ICML) 2024*. [\[Link\]](#)
5. Jiashuo Liu⁺, Tianyu Wang⁺, Peng Cui, Hongseok Namkoong. On the Need for a Language Describing Distribution Shifts: Illustrations on Tabular Datasets. *Neural Information Processing Systems (NeurIPS) 2023, Datasets and Benchmarks Track*. [\[Link\]](#)
 - **Highlighted as NeurIPS 2023 Favorite Papers by Two Sigma (9/3500+)**. [\[Link\]](#)
6. Garud Iyengar*, Henry Lam*, Tianyu Wang*. Hedging against Complexity: Distributionally Robust Optimization with Parametric Approximation. *Artificial Intelligence and Statistics (AISTATS) 2023*. [\[Link\]](#)
 - **Notable Paper (Oral Presentation), 32/1689 = 1.9% of submissions.**

WORKING PAPERS

1. Henry Lam*, Tianyu Wang*. Achieving First-order Statistical Improvements in Data-Driven Optimization. *To be submitted to Operations Research*. Preliminary version appeared in NeurIPS 2025 MLxOR Workshop and OPT Workshop on Optimization for Machine Learning.
2. Tianyu Wang, Naz Pinar Taskiran, Garud Iyengar. Optimizing Pharmaceutical Control with Multi-Task Contextual Bandits: Addressing Batch Heterogeneity for Improved Manufacturing Efficiency. *Major revision at Manufacturing & Service Operations Management*. [\[Link\]](#)
 - **Finalist of MSOM Data-Driven Research Challenge 2025.**
3. Jiashuo Liu⁺, Tianyu Wang⁺, Henry Lam, Hongseok Namkoong, Jose Blanchet. DRO: A Python Library for Distributionally Robust Optimization in Machine Learning. *R&R (aka Major revision) at Journal of Machine Learning Research*. Preliminary version appeared in NeurIPS 2025 Optimization for Machine Learning Workshop. [\[Link\]](#) [\[Website\]](#) [\[Python Package\]](#)
4. Garud Iyengar*, Henry Lam*, Tianyu Wang*. Hedging Complexity in Generalization via a Parametric Distributionally Robust Optimization Framework. *Major revision at Management Science*. Preliminary version appeared in AISTATS 2023. [\[Link\]](#)
5. Tianyu Wang, Ningyuan Chen, Chun Wang. Contextual Optimization under Covariate Shift: A Robust Approach via Intersecting Wasserstein Ambiguity Balls. *Major revision at Manufacturing & Service Operations Management*. [\[Link\]](#)

AWARDS AND HONORS

RESEARCH

- Finalist, NUS Next-Gen Scholar Symposium 2025
- Deming Fellowship, Columbia Business School 2025
- Honorable Mention of Dupačová-Prékopa Best Student Paper Prize in Stochastic Programming 2025
- Finalist, MSOM Data-Driven Research Challenge 2025
- Rising star, ISyE-MS&E-IOE Joint Workshop 2025
- NeurIPS Scholar Award 2024
- AISTATS Notable Paper (Oral Presentation) 2023
- Columbia IEOR Department Fellowship 2021

SERVICE

- INFORMS APS Travel Grant 2025
- AISTATS Best Reviewer Award 2025

PRE-DOCTORAL AWARDS

- Outstanding Undergraduate in Tsinghua (*2% in Tsinghua*) 2021
- Distinguished Undergraduate Thesis Award (*6 in Tsinghua SEM*) 2021
- Chen Daisun Scholarship (*3 in Tsinghua SEM*) 2021
- National Scholarship (*0.2% in China*) 2020

TEACHING EXPERIENCE

COLUMBIA UNIVERSITY

Business Analytics (IEOR 4650)

Spring 2022 and Spring 2024

- Head TA for MSc Business Analytics course; held office hours, delivered lectures on basic machine learning models, prepared exam questions (coding in R and Python) and evaluated group projects.
- Teaching score: 4.67/5 (Enrollment: 49, Spring 2022) and 4.75/5 (Enrollment: 29, Spring 2024).

Optimization Models and Methods (IEOR 4004)

Spring 2023

- Head TA for MSc Operations Research core course; held office hours, answered student questions, delivered lectures on optimization solvers, prepared and graded exams.
- Teaching score: 4.34/5 (Enrollment: 110, Spring 2023).

TSINGHUA UNIVERSITY

Foundations of Mathematics and Computer Science

Fall 2018 - Spring 2021

- Tutor for academic and problem-solving guidance in foundation courses such as *Calculus*, *Linear Algebra*, *Probability and Statistics*, and *Computer Programming* for junior students.
- Tutor score: 4.99/5 (Service duration: over 300 hours)

TALKS

Conference or Workshop Presentations:

- INFORMS Optimization Conference, Atlanta, GA March 2026
- BIRS Workshop: Contextual Stochastic Optimization, Banff, Canada February 2026
- INFORMS Annual Meeting, Atlanta, GA October 2025
- Berkeley-Columbia Meeting in Engineering and Statistics, Berkeley, CA October 2025
- NUS Next-Gen Scholar's Symposium September 2025
- Purdue Operations Conference, West Lafayette, IN August 2025
- International Conference on Stochastic Programming (ICSP), Paris, France July 2025
- INFORMS APS Conference, Atlanta, GA June 2025
- INFORMS MSOM Conference, London, UK June 2025
- INFORMS Annual Meeting, Seattle, WA October 2024
- INFORMS Data Science Workshop, Seattle, WA October 2024
- Modeling and Optimization: Theory and Applications (MOPTA), Bethlehem, PA August 2024
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada July 2024
- International Conference of the Chinese Scholars Association for Management Science and Engineering (CSAMSE), Xiamen, China July 2024
- Columbia Foundations of Data Science Workshop, New York, NY April 2024
- Citadel PhD Summit, Miami, FL April 2024
- INFORMS Annual Meeting, Phoenix, AZ October 2023
- International Conference on Artificial Intelligence and Statistics, Valencia, Spain April 2023
- INFORMS Annual Meeting (Virtual) October 2021

Invited Seminars:

- Rutgers Business School, Supply Chain Management February 2026
- Stanford GSB, Operations, Information & Technology January 2026
- U of Utah David Eccles School of Business, Operations and Information Systems December 2025
- National University of Singapore, Industrial Systems Engineering & Management December 2025
- UT Dallas Naveen Jindal School of Management, Operations Management December 2025

- Singapore Management University Lee Kong Chian School of Business November 2025
- HKUST Business School, Information Systems & Operations Management November 2025
- Chinese University of Hong Kong (Shenzhen), School of Data Science November 2025
- Chinese University of Hong Kong, Systems Engineering & Engineering Management November 2025
- ETH Zurich, Department of Management, Technology, and Economics October 2025
- UofT Rotman School of Management, Young Scholar Seminar (Virtual) September 2025
- Tsinghua University, Department of Management Science and Engineering June 2025

PROFESSIONAL SERVICE

REVIEWER

- Journal referee for *Annals of Applied Probability*, *INFORMS Journal on Data Science*, *Journal of Machine Learning Research*, *Operations Research*, *Management Science*.
- Conference referee for *AISTATS*, *ICLR*, *ICML*, *NeurIPS*.

SESSION CHAIR

- INFORMS Annual Meeting, Atlanta, GA October 2025
- INFORMS MSOM Conference, London, UK June 2025
- INFORMS Annual Meeting, Seattle, WA October 2024
- International Symposium on Mathematical Programming (ISMP), Montréal, Canada July 2024

ORGANIZER

- NYC Joint Operations Research PhD Colloquium May 2024

DEPARTMENT SERVICE

- Founding Member of IEOR PhD Council 2023 - 2025

INDUSTRY EXPERIENCE

Amazon Bellevue, WA
Research Scientist Intern Summer 2023
 Project: Uncertainty Attribution in IPC Simulation (return intern offer extended).

Last updated on June 7, 2026.